Research Proposal: LLM-Based Trading Agents in Continuous Double-Auction Markets

1. Background and Motivation

Continuous Double Auctions (CDAs), where buyers and sellers continuously submit orders to a central order book, serve as the backbone for real-world equity and commodity markets. Various open-source simulators facilitate CDA research, ranging from the public-domain Bristol Stock Exchange (BSE) to high-fidelity platforms like ABIDES and PyMarketSim. Recent literature demonstrates the viability of Large Language Models (LLMs) as trading agents within these simulations. Studies indicate that LLMs can accurately follow strategy instructions, emulate roles such as value investors or momentum traders, and generate realistic market behaviors including price discovery and bubbles. Despite these advances, significant areas remain unexplored, including the performance of open-source models (e.g., LLaMA-3), reinforcement learning implementations, advanced memory modules, and multi-agent collusion dynamics.

2. Objectives and Scope

This research aims to systematically evaluate heterogeneous LLM-based trading agents against traditional algorithmic benchmarks within a structured CDA environment. By leveraging structured prompts, memory buffers, and controlled communication protocols, the study will isolate the specific variables that contribute to trading efficacy and market stability. The research will exclusively utilize the ABIDES-based discrete-event simulator due to its support for pairwise latency and message-based exchange communication.

3. Agent Design and Architecture

- **LLM Integration:** Agents will be powered by a mix of proprietary models (GPT-4/Turbo/4o) via OpenAI APIs and open-source models (LLaMA-3, Claude) via Hugging Face to evaluate how model choice influences trading strategy.
- **Role Specification:** System prompts will define specific agent archetypes, such as value investors utilizing dividend discount models, momentum traders, and market-makers.
- **Structured Outputs:** Agents will generate a two-tier output consisting of natural-language reasoning alongside machine-readable JSON orders detailing action, quantity, and price.
- **Memory Buffers:** LLM agents will utilize memory modules to store past outcomes, allowing context such as previous purchase prices to be included in subsequent prompts.

4. Experimental Methodology

All experiments will be conducted locally in Python using ABIDES-Gym as a convenience wrapper for structured, repeated runs. To ensure a constant market size, each simulation run will consist of exactly 10 agents. The following eight controlled experiments will be executed:

- **Experiment 1: Baseline Market Reproduction:** The simulation will run with 10 traditional agents (a mix of GDX, AA, and simple liquidity providers/takers) to verify that the environment produces sensible bid-ask spreads, price discovery, and volume.

- **Experiment 2: Single LLM Trader vs. Traditional Background:** One LLM trader will be inserted into a market with 9 traditional agents to compare profit, fill rate, quote aggressiveness, and overall market impact.
- **Experiment 3: Prompt Ablation:** Using the 1-LLM and 9-traditional setup, the LLM's prompt format will be varied between a minimal prompt, a reasoning prompt, and a structured JSON prompt to determine if explicit reasoning improves behavior or merely increases latency.
- **Experiment 4: Memory Ablation:** The historical information provided to the single LLM will be restricted to test adaptation, varying between no history, the last 5 events, the last 20 events, and a rolling session summary.
- **Experiment 5: Information-Access Ablation:** The market view provided to the LLM will be altered between top-of-book only, full visible limit-order-book, limit-order-book plus recent trades, and limit-order-book plus inventory/recent PnL to isolate the effect of information richness.
- **Experiment 6: Latency Sensitivity:** Artificial delays will be added to the LLM's decision process (no delay, small delay, moderate delay, large delay) to evaluate competitiveness when reaction-speed advantages are lost.
- **Experiment 7: Shock-Response Market:** A sudden change in fundamental value or supply-demand regime will be introduced mid-session to measure the LLM's adaptation speed, profit loss during the shock, and quote re-centering time.
- **Experiment 8: Communication and Collusion:** The market composition will shift to 2 LLM traders and 8 traditional agents, placing both LLMs on the same side of the market to test inter-agent coordination. The ability to exchange private messages before quoting will be varied across three conditions: no communication, restricted communication, and free communication.

5. Evaluation Metrics

- **Financial Performance:** Agents will be evaluated using standard metrics including Cumulative Return (CR), Sharpe Ratio (SR), Maximum Drawdown (MDD), and overall profit and loss.
- **Market Dynamics:** Outcomes such as price efficiency (convergence to fundamental value), bid-ask spread, market depth, volatility, and the occurrence of bubbles will be recorded.
- **Collusion Indicators:** For Experiment 8, specific anti-competitive metrics will be measured, including spread widening, undercutting frequency, quote synchronization, and seller-side surplus.
- **Qualitative Analysis:** LLM reasoning outputs will be manually inspected to verify strict adherence to the assigned trading strategies.